

## Linear Algebra – Problem Set 4

*Vector Spaces & Bases · 12 concept-first problems · Original problems in the spirit of MIT 18.06 (Strang)*

### Problem 1

Test each set for being a **subspace** of  $\mathbb{R}^2$ , using the three-step test: (i) does it contain the zero vector, (ii) is it closed under addition, (iii) is it closed under scalar multiplication?

### Problem 2

Which of the following are vector spaces over  $\mathbb{R}$ ? For those that are not, name the first axiom that fails (and, if it is a subspace of something, say what subspace it *is*).

### Problem 3

This set is about meaning, so here is a set about *why the axioms say what they say*. For each axiom below, explain in one or two sentences what it buys you — that is, what would go wrong in the rest of linear algebra if it were dropped.

### Problem 4

Let  $S$  be any set of vectors.

### Problem 5

Three descriptions of a basis appear in textbooks:

### Problem 6

Describe the geometry of the subspaces of  $\mathbb{R}^3$  — again, all through the origin.

### Problem 7

Let

### Problem 8

In Set 3 we used two bases of  $\mathbb{R}^3$ :

### Problem 9

For each of the following linear maps  $\mathbb{R}^2 \rightarrow \mathbb{R}^2$ , find a basis of the kernel and a basis of the image, and decide whether the map is injective and whether it is surjective:

### Problem 10

Let  $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$  be  $T(x, y, z) = (x + y, y + z)$ .

### Problem 11

Prove the **rank–nullity theorem**: for a linear map  $T$  from an  $n$ -dimensional space to any space,

### Problem 12

True or false? For the false ones, give a counterexample; for the ones that depend on a convention, say which convention makes them true and why.